Jack Basinski

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Education

University of Washington — Seattle, WA

Bachelor of Science in Electrical and Computer Engineering — *September 2023 – Present*

- GPA: 3.7, College Honors student, Dean's List scholar.
- **Relevant Coursework**: Analog CMOS Integrated Circuits, Applied Electromagnetics, Antenna Design, Computer Architecture, Semiconductor Device Physics, Design of Digital Circuits and Systems.

Experience

Photolithography Process Engineer Intern — June 2026 – September 2026

Micron Technology — *Boise, ID*

- Supported DRAM photolithography process development through SPC analysis, process characterization, and manufacturing data analysis.
- Consolidated a two-step photo coating flow (SOG/SOC) into a single MLR process, enabling **400 hr/week cycle-time reduction** and improving coat-tool capacity across the DRAM line.

Undergraduate Research Assistant — May 2025 – Present

Washington Nanofabrication Facility — *Seattle, WA*

- Execute semiconductor manufacturing processes to **serve 150+ academic and commercial facility users.**
- Characterized photoresist spin curves supporting process qualification and manufacturing operations.
- Enforce ISO 5 cleanroom operational integrity.

Lab Assistant — September 2024 – January 2026

UW Nanodevice Physics Group — *Seattle, WA*

- Assemble nanodevices using VDW materials for research in energy, information, and sensing applications.
- Fabricated a 16-channel feedback lock-in amplifier for efficient multi-terminal transport measurements. Enabled parallelization of data collection, **eliminating several thousand dollars in lab expenses.**

Projects

CMOS 5T Operational Transconductance Amplifier | *Cadence Virtuoso* — May 2026 – June 2026

- Designed, simulated, and optimized a CMOS 5T OTA in **Cadence Virtuoso (GF gpdk045)** using sensitivity analysis and transistor-level design optimization.
- Achieved **22.2 dB gain, 1.40 GHz GBW, 108 MHz bandwidth** while meeting stability requirements.

Adaptive Interference Cancellation System | *Python, Arduino, Analog Circuits* — January 2026 – March 2026

- Designed a real-time interference cancellation system with FFT-based frequency detection and automatic filter tuning using Python, Arduino, and mixed-signal hardware.
- Achieved adaptive interference suppression across the **2–20 kHz** band with **sub-kHz tuning accuracy**.

4×1 Microstrip Patch Antenna Array | *Ansys HFSS, Altium Designer* — January 2026 – March 2026

- Designed, simulated, fabricated, and validated a 5.8 GHz 4×1 microstrip patch antenna array on FR-4 substrate, meeting all specifications with **9.18 dBi peak gain and fan-beam radiation pattern.**
- Extended design with a stacked parasitic-patch configuration, **expanding bandwidth 5× from 258 MHz to 1.3 GHz while improving peak gain to 11.1 dBi.**

Pipelined CPU | *SystemVerilog, Assembly, Modelsim* — June 2025 – August 2025

- Designed a 5-stage pipelined CPU around ARM's LEGV8 instruction set. Simulated and validated performance across benchmark algorithms, demonstrated **x2.7 speedup over non-pipelined implementation.**
- Implemented forwarding hardware solutions, eliminating data hazards and **improving CPI by 70%**

Technical Skills

Languages: SystemVerilog, Assembly, Java, Python, C, MATLAB.
Engineering Tools: Cadence EDA, Nanofabrication Process Tools (DELM), FPGA Development, Modelsim, Ansys HFSS, Altium Designer, Microcontrollers, Analog + Logic Circuits, Soldering, SolidWorks, Microsoft Excel.
General Knowledge + Interests: Mandarin Chinese, Photography, Hiking, Pearl Jam.